Manoj Reddy Manchala

Manoj Manchala | Manoj Reddy Manchala | applications.manojreddy@gmail.com | +91 7780611965

Education

University of California San Diego **Sept 2023 - Jun 2025**
- Master of Science in Mechanical Engineering (Specialized in **Robotics**)
- **Key Courses:** Planning & Learning Robotics, Sensing & Estimation in Robotics, Safety for Autonomous Systems

Indian Institute of Technology, Bombay **July 2017 - May 2021**
- Bachelor of Technology in Mechanical Engineering

Technical Skills

Programming	Python, ROS, Linux, Pandas, Numpy, OpenCV, Machine Learning, MATLAB, R, LaTeX
Tools/Frameworks	MuJoCo, PyTorch, JAX, GTSAM, CUDA, ONNX, TensorRT, Git, TensorFlow, Simulink
Specialized Skills	Sensor Fusion, Computer Vision, Perception, Object Detection, Deep Learning

Experience

Graduate Student Researcher - Existential Robotics Laboratory **Mar 2024 - Oct 2025**

Enhanced the Transformer-based perception framework with RGB data enabling **semantic awareness** of the agent, combining perception, manipulation, and navigation into a **unified Reinforcement Learning policy**.

Cross-Modal Transformer for Temporal Context:
- Architected a cross-modal Vision Transformer integrating CNN-based **visual patch embeddings** & MLP-encoded state tokens within a **Multi-head Transformer Encoder** to generate temporally aware state representations for RL policies.
- Outperformed the GNN with **20%** increase in mean velocity, **135%** increase in average coverage, & a **27%** reduction in collision frequency - attributed to Transformer's ability to capture spatiotemporal dependencies

GNNs for Enhanced Spatial Understanding:
- Implemented a **graph neural network** (GNN)-based feature extractor, modeling **point clouds as structured graphs** to enhance 3D object classification and spatial reasoning, significantly improving scene understanding.
- Achieved a **21% increase** in navigation speed, **85% increase in distance covered** & **17% reduction in collisions**.

Vision Guided Autonomous Navigation:
- Engineered a Deep Reinforcement Learning framework based on policy-gradient methods (**PPO/SAC**) for autonomous navigation of an Ackermann-drive vehicle, enabling efficient maneuvering in complex, obstacle-rich environments.
- Developed a physics-based custom simulation environment with **MuJoCo/DM Control** for realistic training.
- Converted PyTorch models to **ONNX** & **TensorRT** for deployment on **NVIDIA Jetson TX2** (F1TENTH car).

Business Leadership Trainee - FARE Labs Private Limited **Jul 2021 - Feb 2022**
- Created a repository of segment segregated clients through detailed Market Research, optimizing targeted outreach.
- Executed innovative marketing strategies, **boosting revenue by 15%**, securing clients worth **$250k**

Research Internship - Varroc Tech Center **Nov 2019 - Dec 2019**
- Simulated & Evaluated multiple state-of-charge estimation techniques for Li-ion batteries, prioritizing model-based approaches like PI observers, **Sliding Mode observers**, and **Kalman filters**.

Projects

Localization and Mapping :
- Developed a **Visual-Inertial SLAM** system via an Extended Kalman Filter (EKF) prediction, integrating IMU & stereo camera data to accurately estimate the robot's trajectory & landmarks.
- Crafted a LiDAR-based SLAM system for differential-drive bot, performed point cloud registration via **Iterative Closest Point** & **Pose Graph Optimization** enhancing trajectory accuracy, generating detailed occupancy & texture maps.

Motion Planning :
- Formulated a **dynamic programming** approach for a 'Door & Key' navigation, formulating the task as a Markov Decision Process, demonstrating effective path planning for both known and random maps.
- Implemented **A*** & **RRT** to solve navigating through various complex 3D environments with static obstacles.
- Engineered a trajectory tracking control system for a differential-drive robot using **Generalized Policy Iteration** & **Certainty Equivalent Control** with minimal trajectory deviations while avoiding obstacles.

MPC for Uneven Terrains :
- Engineered a **Model Predictive Control** formulation for ground robots, enabling safe navigation on complex terrains with friction and elevation variations based on a customized dubins car model.

Comma.ai Calibration Challenge :
- Devised a **Conv-LSTM network** integrating **optical flow** features with RGB images to predict vehicle pitch & yaw from monocular camera inputs, achieving **MSE of 21.72%** & robust to varying environmental conditions.

Mobile Manipulator :
- Developed a control system for a mobile manipulator robot in MATLAB & **CoppeliaSim**, implementing kinematics simulation, **trajectory generation**, PI control strategy for precise autonomous navigation & **object manipulation**.